

Nonexpansive Mappings

Chapter 3b: Nonlinear Analysis and Fixed Point Theory

Based on H.K. Pathak (2018)

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An Introduction to Nonlinear Analysis and Fixed Point Theory

Outline

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What are Nonexpansive Mappings?

Definition

A mapping $T : X \rightarrow X$ on a Banach space X is called **nonexpansive** if

$$\|Tx - Ty\| \leq \|x - y\| \quad \forall x, y \in X$$

Key Properties

- Lipschitz continuous with constant $L = 1$
- Preserves distances (contractions use $L < 1$)
- Generalizes the concept of isometries (when $L = 1$)
- Fundamental in variational analysis and optimization

Why Important?

- Naturally arise in optimization and PDE theory
- Weaker condition than contractive mappings
- Rich fixed point theory despite weaker Lipschitz constant

Historical Context

1960s Introduction of nonexpansive mappings in fixed point theory

Browder, Kirk Developed surjectivity and fixed point theorems

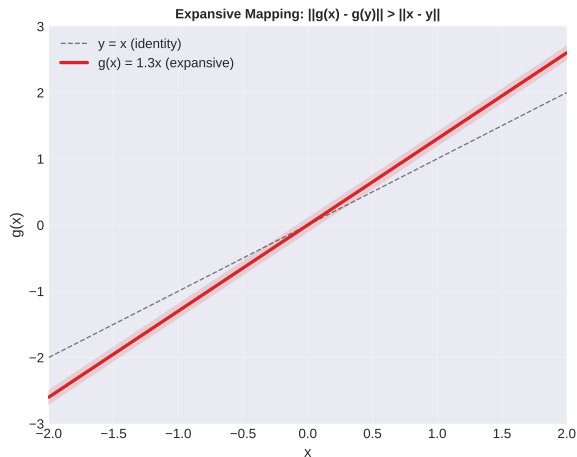
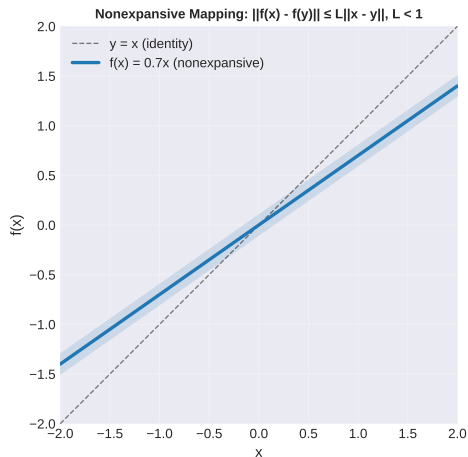
Connections Links to monotone operators, optimization, and convex analysis

Modern Applications Machine learning, variational inequalities, PDE solutions

Classical Problem: Does every nonexpansive mapping on a compact convex set in a Hilbert space have a fixed point?

Answer (Browder, 1965): Yes! For nonexpansive mappings in reflexive Banach spaces.

Nonexpansive vs Other Mappings I



Contractive $\|Tx - Ty\| \leq k\|x - y\|$ with $k < 1$ (always have fixed points)

Nonexpansive vs Other Mappings II

Nonexpansive $\|Tx - Ty\| \leq \|x - y\|$ (Lipschitz constant = 1)

Expansive $\|Tx - Ty\| > \|x - y\|$ (may not have fixed points)

Nonexpansive Sequences

Definition

Let E be a Banach space. A sequence $\{x_n\}_{n \geq 0}$ in E is called:

- 1 **Nonexpansive sequence:** if

$$\|x_{n+1} - x_n\| \leq \|x_n - x_{n-1}\| \quad \forall n \geq 1$$

- 2 **Generalized nonexpansive sequence:** if it satisfies the above condition

- 3 **Almost nonexpansive sequence:** if

$$\|x_{n+1} - x_n\| \leq \|x_n - x_{n-1}\| + \varepsilon(i, j)$$

where $\{\varepsilon(i, j)\}_{i, j \geq 0}$ is bounded and $\lim_{i \rightarrow \infty} \varepsilon(i, j) = 0$

Key Definitions in Nonexpansive Theory

Definition (Lipschitz Continuity)

A mapping $f : X \rightarrow Y$ is **Lipschitz continuous** if there exists $L \geq 0$ such that

$$\|f(x) - f(y)\| \leq L\|x - y\| \quad \forall x, y \in X$$

The smallest such L is called the **Lipschitz constant**.

Definition (Fixed Point)

A point $x^* \in X$ is a **fixed point** of T if $Tx^* = x^*$.

$$\text{Fix}(T) = \{x \in X : Tx = x\}$$

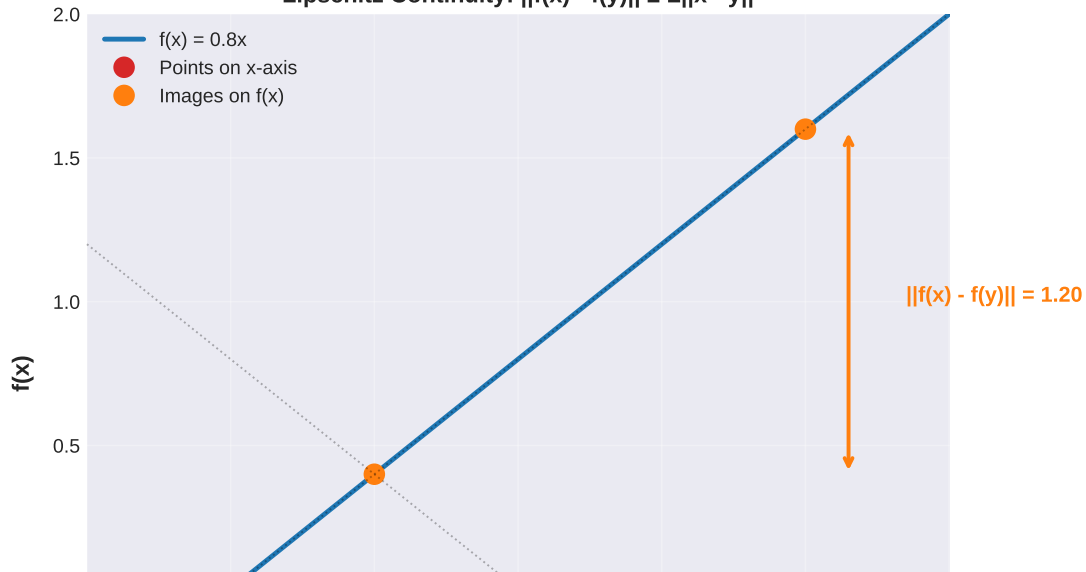
Definition (Asymptotic Behavior)

For a sequence $\{x_n\}$, the **asymptotic center** K is the set of all weak cluster points of subsequences

Lipschitz Constant Illustration I

Lipschitz Constant Illustration II

Lipschitz Continuity: $\|f(x) - f(y)\| \leq L\|x - y\|$



Fundamental Properties

Let $T : X \rightarrow X$ be a nonexpansive mapping on a Banach space X .

- ① If T has a fixed point x^* , then $Tx = x$ implies $x = x^*$ (uniqueness under contraction)
- ② The composition of nonexpansive mappings is nonexpansive
- ③ For a nonexpansive sequence $\{x_n\}$:

$$\|x_m - x_n\| \leq \|x_m - x_{m-1}\| + \|x_{m-1} - x_{m-2}\| + \cdots + \|x_{n+1} - x_n\|$$

- ④ Differences form a Cauchy sequence if $\sum_{n=1}^{\infty} \|x_{n+1} - x_n\| < \infty$

Convergence in Nonexpansive Sequences

Lemma 3.4

Let $\{a_n\}_{n \geq 1}$ be a sequence of positive reals with $b_n = \sum_{i=1}^n a_i$ and $b_n \uparrow \sum_{i=1}^{\infty} a_i = \infty$. If $\{x_n\}$ is a sequence of reals with $x_n \rightarrow x$, then

$$\lim_{n \rightarrow \infty} \frac{1}{b_n} \sum_{i=1}^n a_i x_i = x$$

Application: This is crucial for establishing mean convergence of nonexpansive sequences.

Lemma 3.5

Let $\{x_n\}_{n \geq 0}$ be a generalized nonexpansive sequence in a reflexive Banach space E . Define

$$K = \bigcap_{n=1}^{\infty} \overline{\text{co}} \left\{ \left[\frac{x_i - x_{i-1}}{i} \right]_{i \geq n} \right\}$$

Then K is nonempty, closed, and convex.

Definition and Properties

Definition (Generalized Nonexpansive Sequence)

A sequence $\{x_n\}_{n \geq 0}$ in Banach space E is **generalized nonexpansive** if

$$\|x_{n+1} - x_n\| \leq \|x_n - x_{n-1}\| \quad \forall n \geq 1$$

This means consecutive differences form a nonexpansive (decreasing) sequence.

Key Observation

- The sequence of differences is monotone decreasing:

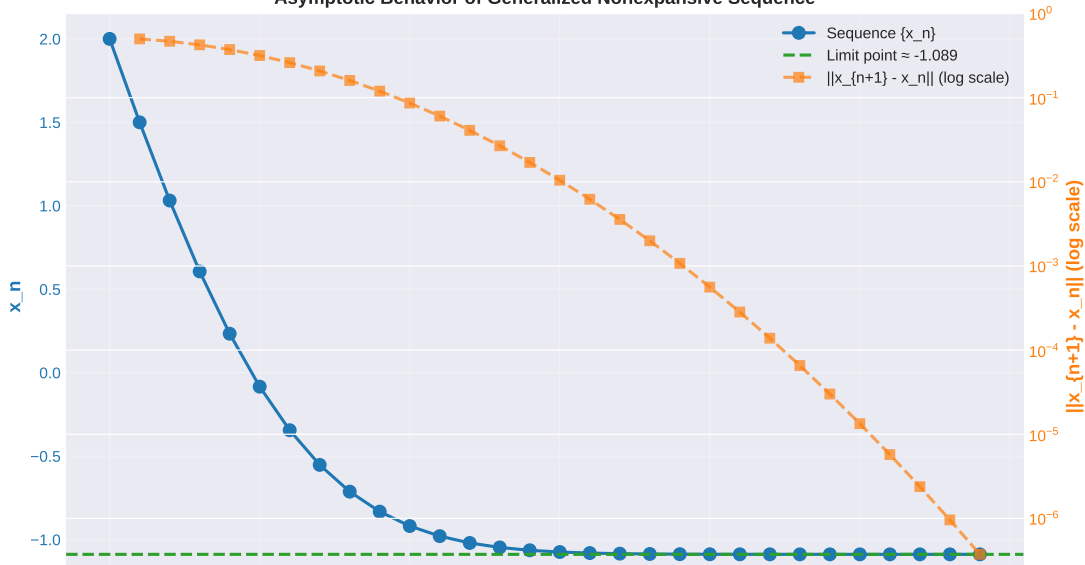
$$\|x_n - x_{n-1}\| \geq \|x_{n+1} - x_n\| \geq \|x_{n+2} - x_{n+1}\| \geq \dots$$

- This implies the differences converge to zero or form a Cauchy sequence
- Provides a natural framework for asymptotic analysis

Convergence Behavior I

Convergence Behavior II

Asymptotic Behavior of Generalized Nonexpansive Sequence



Asymptotic Behavior - Main Result

Theorem (Asymptotic Center - Theorem 3.32)

Let E be a reflexive Banach space and $\{x_n\}$ a generalized nonexpansive sequence in E . Define

$$K = \bigcap_{n=1}^{\infty} \overline{\text{co}} \left\{ [x_i - x_{i-1}]_{i \geq n} \right\}$$

Then

- ① $d = d(0, K) = d\left(0, \overline{\text{co}} \left\{ \frac{x_n - x_0}{n} \right\}\right)$
- ② There exists a unique point $z_0 \in K$ with $\|z_0\| = d$
- ③ $z_0 \in \overline{\text{co}} \left\{ \frac{x_n - x_0}{n} \right\}$ with $\|z_0\| = d$

Interpretation: The asymptotic center characterizes where the normalized differences accumulate.

Corollary 3.5: Special Cases

Corollary 3.5

Under the conditions of Theorem 3.32 with E , $\{x_n\}$, K , and d as above:

① If E is strictly convex:

weak- $\lim_{n \rightarrow \infty} \frac{x_n}{n}$ exists and coincides with $P_K 0$ with $\|P_K 0\| = d$

② If E^* has Fréchet differentiable norm:

strong- $\lim_{n \rightarrow \infty} \frac{x_n}{n}$ exists and coincides with $P_K 0$

Here $P_K 0$ denotes the projection of 0 onto K .

Definition and Generalization

Definition (Almost Nonexpansive Sequence)

A sequence $\{x_n\}_{n \geq 0}$ in a Banach space E is **almost nonexpansive** if

$$\|x_{n+1} - x_n\| \leq \|x_n - x_{n-1}\| + \varepsilon(i, j)$$

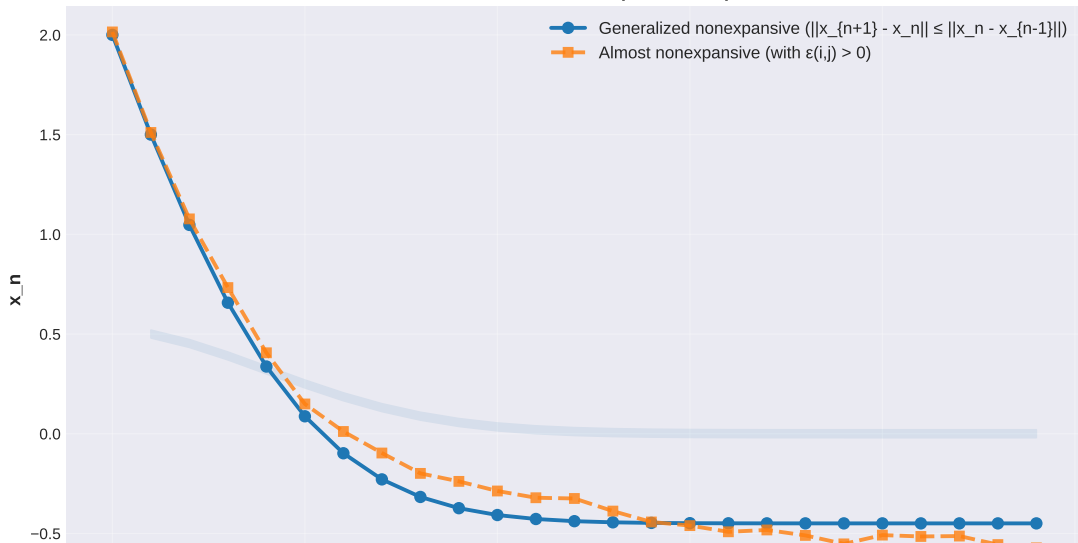
where:

- $\{\varepsilon(i, j)\}_{i, j \geq 0}$ is bounded: $|\varepsilon(i, j)| \leq M$ for all i, j
 - The perturbation vanishes: $\lim_{i \rightarrow \infty} \varepsilon(i, j) = 0$ for each j
-
- Generalizes the pure nonexpansive case (set $\varepsilon(i, j) = 0$)
 - Naturally arises in numerical algorithms with perturbations
 - Includes approximations of fixed point iterations
 - Still retains good convergence properties

Comparison: Pure vs. Almost Nonexpansive I

Comparison: Pure vs. Almost Nonexpansive II

Generalized vs. Almost Nonexpansive Sequences



Theorem 3.33: Application to Inequalities

Theorem (Theorem 3.33)

Let $f_1, f_2 \in L_2([0, 1])$ satisfy

$$\left| \int_0^1 g_1(t) dt \right|^2 - \left| \int_0^1 g_2(t) dt \right|^2 \leq \int_0^1 (f_1(t)g_1(t) + f_2(t)g_2(t)) dt \leq \int_0^1 (|g_1(t)| + |g_2(t)|)^2 dt$$

for all $g_1, g_2 \in L_2([0, 1])$. Then $f_1 = \lambda$ and $f_2 = \mu$ for some $\lambda, \mu \in [-1, 1]$.

Proof Strategy

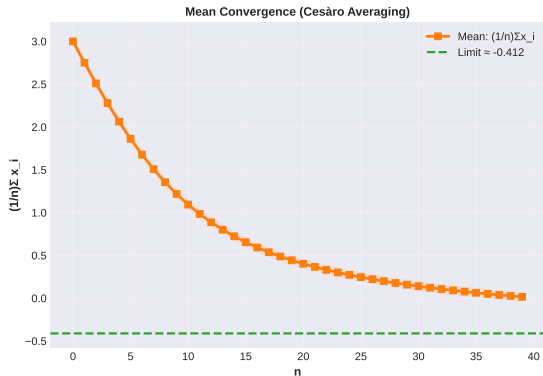
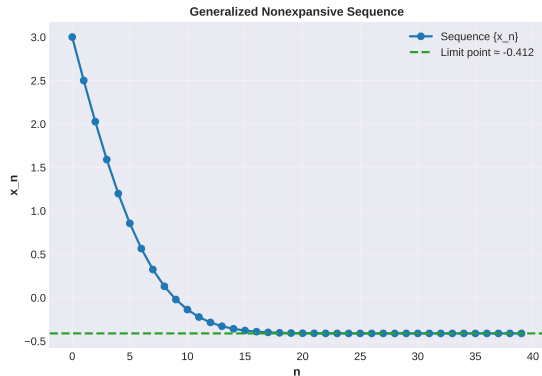
- 1 Define $E = L_2([0, 1]; \mathbb{R}_\infty^2)$ with sup norm
- 2 Consider generalized nonexpansive sequence $x_{2k} = (2k, \frac{1}{2^k})$ and $x_{2k+1} = (2k, \frac{1}{2^{k+1}})$
- 3 Apply Theorem 3.32 to extract the result

Algorithm: Computing Asymptotic Centers

Algorithm for Generalized Nonexpansive Sequences

```
def asymptotic_center(x_seq, tolerance=1e-6):  
    """  
    Compute asymptotic center of generalized nonexpansive sequence.  
  
    Args:  
        x_seq: list of points in the sequence  
        tolerance: convergence tolerance  
  
    Returns:  
        asymptotic_center: the point  $z_0$  with  $\|z_0\| = d$   
        distance:  $d = d(0, K)$   
    """  
    n = len(x_seq)  
    differences = [norm(x_seq[i] - x_seq[i-1])  
                   for i in range(1, n)]  
  
    # Check monotone decrease  
    is_nonexp = all(differences[i] <= differences[i-1]  
                    for i in range(1, len(differences)))  
  
    if not is_nonexp:  
        raise ValueError("Sequence is not nonexpansive")
```

Asymptotic Behavior - Mean Convergence



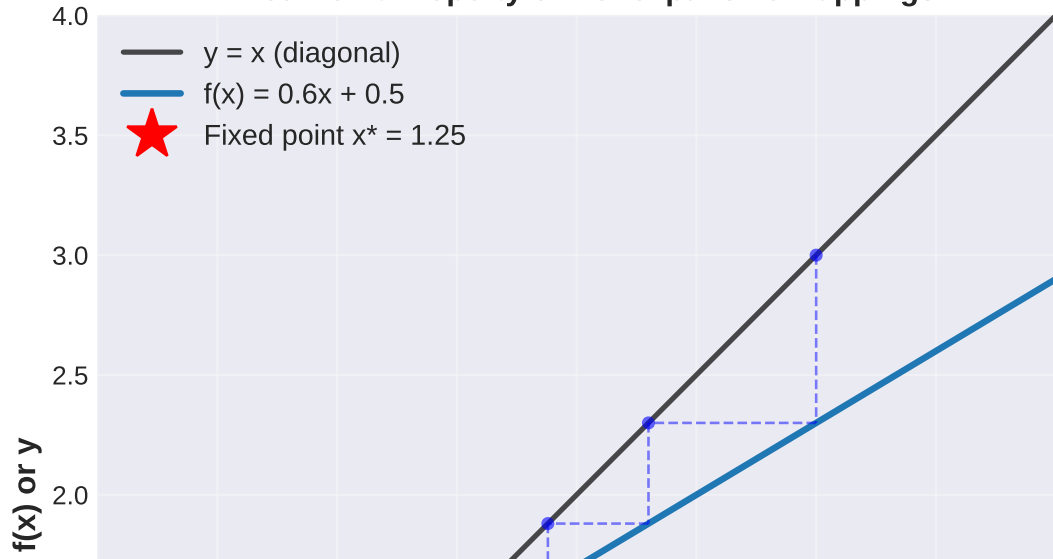
Key Results:

- The Cesàro mean $\frac{1}{n} \sum_{i=1}^n x_i$ converges to the limit
- Convergence is robust: holds even with perturbations
- Rate of convergence depends on the decay of consecutive differences

Fixed Point Property I

Fixed Point Property II

Fixed Point Property of Nonexpansive Mappings



Conditions for Fixed Point Existence

Theorem (Fixed Point Theorems)

A nonexpansive mapping $T : C \rightarrow C$ has a fixed point if any of:

- ① **Browder (1965):** *C is compact and convex in a Hilbert space*
- ② **Kirk (1965):** *C is weakly compact and has the fixed point property for isometries*
- ③ **General:** *C is bounded, closed, convex in a reflexive Banach space AND T is a contraction or has other special properties*

Unlike contractive mappings (Banach Fixed Point Theorem), nonexpansive mappings do NOT always have fixed points without additional hypotheses on the domain or the mapping.

Convergence of Fixed Point Iterations

Convergence Theory

- **Strong Convergence:** Requires strict convexity or other geometric conditions
- **Weak Convergence:** Often achievable for reflexive spaces
- **Cesàro Convergence:** The mean $\frac{1}{n} \sum_{i=0}^{n-1} T x_i$ converges even when x_n doesn't
- **Explicit Convergence:** Can be established using asymptotic analysis of the sequence $\{x_n\}$

Definition (Projection Operator)

The metric projection $P_K : X \rightarrow K$ onto a closed convex set K is the unique point satisfying:

$$P_K(x) = \arg \min_{y \in K} \|x - y\|$$

Relationship to Monotone Operators

Connections

- **Accretive Operators:** Nonexpansive mappings relate to accretive operators in Banach spaces
- **Resolvents:** For maximal monotone operators A , the resolvent $(I + \lambda A)^{-1}$ is nonexpansive
- **Variational Inequalities:** Finding fixed points of nonexpansive mappings is equivalent to solving variational inequalities
- **Optimization:** Proximal mappings in convex optimization are nonexpansive

These topics are developed further in Chapter 4 (Monotone Operators, Strongly ϕ -Accretive Operators and Their Variants).

Applications in Nonlinear Analysis

- **PDEs:** Heat equations, parabolic equations via semigroups
- **Integral Equations:** Hammerstein type equations with nonexpansive nonlinearities
- **Game Theory:** Nash equilibrium problems
- **Machine Learning:** Gradient descent and proximal methods for optimization
- **Dynamical Systems:** Iterative algorithms and their long-time behavior

Example 1: Simple Nonexpansive Mapping

Example

Consider $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = 0.7x + 0.5$.

Verification of Nonexpansivity:

$$|T(x) - T(y)| = |0.7x + 0.5 - 0.7y - 0.5| = 0.7|x - y| \leq |x - y|$$

Lipschitz constant: $L = 0.7 < 1$ (actually contractive)

Fixed Point: Setting $x = T(x)$:

$$x = 0.7x + 0.5 \Rightarrow 0.3x = 0.5 \Rightarrow x^* = \frac{5}{3} \approx 1.667$$

Iteration from $x_0 = 2$:

- $x_1 = 0.7(2) + 0.5 = 1.9$
- $x_2 = 0.7(1.9) + 0.5 \approx 1.83$
- $x_3 \approx 1.776$, then converges toward 1.667

Example 2: Nonexpansive vs. Expansive

Nonexpansive: $T(x) = 0.8x$

- $|T(1) - T(0)| = 0.8 \leq |1 - 0|$ (YES)
- $|T(2) - T(1)| = 0.8 \leq |2 - 1|$ (YES)
- Distances shrink or stay same
- Fixed point: $x^* = 0$ (contraction)

Expansive: $T(x) = 1.5x$

- $|T(1) - T(0)| = 1.5 > |1 - 0|$ (NO)
- $|T(2) - T(1)| = 1.5 > |2 - 1|$ (NO)
- Distances grow
- Iterates diverge from any point

Key Difference: Nonexpansive mappings control growth (Lipschitz constant ≤ 1), ensuring bounded differences and often convergence to fixed points.

Example 3: Generalized Nonexpansive Sequence

Example

Define $\{x_n\}$ by: $x_0 = 2.0$, $x_1 = 1.5$, and for $n \geq 2$:

$$x_{n+1} = x_n + 0.85^{n/5}(x_n - x_{n-1})$$

Check Generalized Nonexpansivity: Since $0 < 0.85 < 1$, we have:

$$\|x_{n+1} - x_n\| = 0.85^{n/5} \|x_n - x_{n-1}\| \leq \|x_n - x_{n-1}\|$$

Sequence Values:

```
n=0: x_n = 2.000, |x_n - x_{n-1}| = N/A
n=1: x_n = 1.500, |x_n - x_{n-1}| = 0.500
n=2: x_n = 1.425, |x_n - x_{n-1}| = 0.075
n=3: x_n = 1.410, |x_n - x_{n-1}| = 0.015
...
n -> infinity: x_n -> 1.4 (approximately)
```

The sequence converges and normalized differences converge to the asymptotic center.

Example 4: Application to Integral Inequality

Example

Problem: Prove the inequality (Theorem 3.33):

$$\left| \int_0^1 g_1(t) dt \right|^2 - \left| \int_0^1 g_2(t) dt \right|^2 \leq \int_0^1 (f_1(t)g_1(t) + f_2(t)g_2(t)) dt$$

Setup:

- Let $E = L_2([0, 1])$ with standard inner product
- Consider test functions g_1, g_2
- Define a generalized nonexpansive sequence in $E \times E$

Solution Strategy: Use Theorem 3.32 to identify f_1 and f_2 as elements of the asymptotic center, achieving the inequality bound.

Key Concepts Summary

- ➊ **Nonexpansive Mappings:** Satisfy $\|Tx - Ty\| \leq \|x - y\|$ (Lipschitz constant $L = 1$)
- ➋ **Generalized Nonexpansive Sequences:** Consecutive differences decrease monotonically
- ➌ **Asymptotic Center:** Characterizes accumulation points of normalized differences (Theorem 3.32)
- ➍ **Almost Nonexpansive:** Tolerate small perturbations $\varepsilon(i, j) \rightarrow 0$ while preserving convergence
- ➎ **Fixed Point Theory:** Nonexpansive mappings have fixed points under certain domain conditions
- ➏ **Convergence Modes:**
 - Strong convergence (requires strict convexity)
 - Weak convergence (reflexive spaces)
 - Cesàro convergence (robust, always available)
- ➐ **Applications:** Monotone operators, variational inequalities, optimization, PDEs

Comparison: Three Classes of Mappings

Property	Contractive	Nonexpansive	Expansive
Lipschitz Constant	$L < 1$	$L = 1$	$L > 1$
Distance Behavior	Shrink	Preserve	Grow
Fixed Point Exists	Always	Conditional	Rarely
Uniqueness of FP	Yes	Depends	No
Iteration Convergence	Always	Conditional	Diverges

Further Reading and References I

Key References from the Text

- Pathak, H. K. (2018). *An Introduction to Nonlinear Analysis and Fixed Point Theory*
- Browder, F. E. (1965). Fixed point theorems for noncompact mappings
- Kirk, W. A. (1965). A fixed point theorem for mappings which do not increase distances
- Kachurovskii, R. I. (1960). Monotonic operators and convex functionals

Topics in Later Chapters

- Chapter 4: Monotone Operators and ϕ -Accretive Operators
- Chapter 5: Approximation Methods and Hybrid Algorithms
- Generalizations: Quasi-nonexpansive, asymptotically nonexpansive mappings

Exercises for Practice

- ④ Verify that the following mappings are nonexpansive:
 - ① $T(x) = \begin{cases} x/2 & \text{if } |x| \leq 1 \\ 0 & \text{otherwise} \end{cases}$ on $\mathbb{R}$
 - ② Projection onto a closed convex set in a Hilbert space
- ② Prove that for a nonexpansive sequence, if $\sum_{n=1}^{\infty} \|x_{n+1} - x_n\|^2 < \infty$, then $\{x_n\}$ converges.
- ③ For the sequence $x_0 = 1$, $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ (Newton's method for $\sqrt{2}$), show it is generalized nonexpansive and find its limit.
- ④ Given the iteration map $T(x) = \frac{x+1}{2}$, determine:
 - ① Whether T is nonexpansive
 - ② All fixed points
 - ③ Convergence behavior from initial point $x_0 = 0$

Thank You!

Questions?

For more information on nonexpansive mappings and fixed point theory,
see Pathak (2018) Chapter 3b and Chapter 4.